FAEZE (FAE) FATHIJAM

+1 (814) 769 0441 fkf5073@psu.edu linkedin.com/in/faeze-fathi-jam State College, PA

EDUCATION

Pennsylvania State University, State College, PA 2023 - Present
Ph.D. in Transportation Engineering with Minor in Computational Science
- GPA: 3.89/4

Sharif University of Technology, Tehran, Iran 2019 - 2022
M.Sc. in Highway and Transportation Engineering
- GPA: 3.37/4 (15.66/20)

Malayer University, Hamedan, Iran 2012 - 2016
B.Sc. in Civil Engineering
- GPA: 2.91/4 (15.23/20)

SKILLS AND TECHNIQUES

Programming Language: Python, R, Matlab
Techniques: Machine Learning; Deep Learning; Data Visualization; Numerical Analysis; Bayesian Statistics; Choice Modeling; Spatial Analysis; Time Series Analysis; Econometrics
Software: ArcGIS, AutoCAD Civil3D, Aimsun, Synchro, PTV Vissim, GAMS, Lingo, SPSS, AnyLogic, Arena

RESEARCH EXPERIENCE

Pennsylvania State University 2024 - 2025

Lane-Level Traffic Forecasting with Sequential Graph-based Deep Learning

Built deep learning models (LSTM, CNN-LSTM, Graph Neural Networks) on PeMS lane-level data to jointly forecast flow, speed, and occupancy, demonstrating superior performance of graph-temporal models for capturing merging and spatial dependencies.

Pennsylvania State University 2023 - 2024

Machine Learning for Mixed Car–Bike Traffic Prediction

Developed and evaluated machine learning models on synthetic traffic data to predict vehicle–bicycle interactions in mixed traffic, using classification and regression methods to assess flow, occupancy, and safety implications.

Sharif University of Technology 2019 - 2022

Deep Learning for Railway Track Degradation Prediction

Applied CNN, LSTM, and CNN-LSTM models (Keras, Scikit-Learn) with optimized sequence lengths of 3–6 on 14,000 km of Iranian Railway EM120 geometry time-series data, including preprocessing and evaluation using mean square error (MSE) and R-square (R^2).

PUBLICATIONS

Fathijam, F., Guler, S.I., “Machine Learning-Based Traffic State Prediction in Car-Bike Mixed Traffic using Synthetic Data,” TRB 105th Annual Meeting, Washington, D.C., 2026.

Fathijam, F., Guler, S.I., “Lane-Level Traffic State Prediction Using PeMS Caltrans Dataset,” TRB 105th Annual Meeting, Washington, D.C., 2026.

Fathijam, F., Shafahi, Y., “*Application of Deep Learning Algorithms in Railway Track Degradation Modelling*,” The Fifth International Conference on Railway Technology, Aug. 22-25, 2022, Montpellier, France.

Lu, M., Liu, H., Fathijam, F., Guler, S.I., “Impact of Bikes on Traffic Efficiency Based on Car-Bike Mixed Traffic Flow Modeling,” (Under Review)

Fathijam, F., Ranjbari, A., Akalın, K.B., Guler, S.I., "Why Wouldn't you Walk/Bike? An Analysis of Modal Choices for Non-Work Trips within a Walkable Distance," (Under Review)

TEACHING EXPERIENCE

Pennsylvania State University 2025 - Present

Teaching Assistant for CE 310 Surveying *Instructor: Brian Naberezny*

Taught the lab component, guiding students in CAD and Civil 3D through project-based learning and hands-on field labs with surveying equipment (total station, automatic level)

Pennsylvania State University 2024 - 2025

Teaching Assistant for CE 310 Surveying *Instructor: Brian Naberezny*

Pennsylvania State University 2024

Teaching Assistant for CE 321 Highway Engineering *Instructor: Ilgin Guler*

LEADERSHIP AND INVOLVEMENT

Treasurer, *Institute of Transportation Engineers (ITE) student chapter, Penn State*
2024 - 2025

- Coordinated funds and travel for students attending the Transportation Research Board annual meeting

PROFESSIONAL EXPERIENCE

Steel Detailer 2016 - 2018

Batab Sanat Paydar, Tehran, Iran

- Modeled and detailed steel and concrete buildings using Tekla Structures software and prepared the corresponding technical reports.

Teacher, Educational Counselor Jun. 2014 – Sep. 2014

Kanoon Farhangi Amouzesh Ghalamchi, Tehran, Iran

- Taught Mathematics to middle school students and provided academic guidance.

SELECTED COURSES AND COURSE PROJECTS

- **Traffic Flow Theory; Traffic Operations; Transportation Safety**
- **Transportation Network & Systems Analysis; Transportation Planning; Transportation Demand Analysis**
- **Applied Statistics; Data-Driven Design**
- **Investigated the impact of emergency response time on crash severity** using **Pennsylvania statewide crash data**, accounting for roadway, traffic, and crash characteristics.
- **Calculated on-ramp junction capacity** of a freeway for "**Advanced Traffic Engineering**" course project.
- **Developed an agent-based simulation model** in Anylogic software for optimizing a commodity distribution system for "**Simulation**" course project.
- **Developed queue and inventory simulation programs** in Matlab and comparing results with the same programs created in ARENA software for "**Simulation**" course project.
- **Assessed pavement management systems** of different states in the USA for "**Pavement Management System**" course project.
- **Modeled geometric, and pavement design** of a 20 km road using AutoCAD Civil 3D software for "**Geometric Design**" course project.

REFERENCES

Ilgin Guler, Associate Professor, Department of Civil Engineering, Pennsylvania State University, State College, PA. Email: sig123@psu.edu

Yousef Shafahi, Professor, Department of Civil Engineering, Sharif University of Technology, Tehran, Iran. Email: shafahi@sharif.edu